

Automated Single-Stranded DNA (ssDNA) Library Preparation

Workflow protocol – Hamilton NGS STAR platform, MPI-EVA (“MPI_ssDNA_LibraryPreparation” method)

1. Purpose and background

This document describes, at the workflow level, the automated single-stranded DNA (ssDNA) sequencing-library preparation method that runs on a Hamilton NGS STAR liquid-handling platform at the Max Planck Institute for Evolutionary Anthropology (MPI-EVA). It was compiled from the Hamilton Venus method files for “MPI_ssDNA_LibraryPreparation” (the method's script, subroutine, step, and compiled method files).

The method automates a splinted-ligation ssDNA library preparation approach developed for highly degraded and ancient DNA, of the type described by Gansauge and Meyer. Unlike standard double-stranded library preparation, adapters are added to single-stranded template molecules sequentially, with the partially built library immobilized on streptavidin beads between steps rather than through blunt-end ligation of both strands at once. This makes the approach well suited to short, damaged, or nicked template molecules that would be lost in conventional double-stranded workflows.

In outline, the robot: (1) removes uracils and dephosphorylates the 3' ends of the input DNA, (2) splint-ligates a biotinylated first adapter onto each template strand, (3) captures the ligated molecules on streptavidin beads, (4) synthesizes the complementary second strand by primer extension, (5) ligates a second adapter onto the new strand, and (6) heat-elutes the finished, double-adapter library from the beads and dilutes it for downstream indexing PCR.

2. Scope

The method supports batch sizes from 8 to 96 samples per run (a separate operator input worklist exists for each batch size), and each of the six processing steps (Process 1 through Process 6) can be individually enabled, skipped, or designated as a mandatory stop point. This allows partial runs — for example, stopping and storing the plate at 4 °C after bead immobilization (Process 3) and continuing on a later day. Optional USER (uracil-DNA excision) enzyme treatment can be added alongside the standard dephosphorylation step for damage-repair workflows on ancient or degraded samples.

3. Platform and deck configuration

The method runs on a Hamilton NGS STAR deck fitted with: independent 8-channel pipetting arms configured for low- and standard-volume tips (50/300/1000 µl); a Multi-Probe Head (MPH) for plate-to-plate reagent transfers; an on-deck thermal cycler (ODTC) used for heat-denaturation, dephosphorylation, ligation-inactivation, fill-in, and elution incubations; several CPAC cooled positions that hold sample and reagent plates at 4 °C between steps; Hamilton Heater-Shaker (HHS) units used for room-temperature, 37 °C, and 45 °C incubations with intermittent shaking; an on-deck

magnetic separation station for bead-capture steps; and a stacker/Cytomat that automatically replenishes tips and plates during the run.

4. Reagents and master mixes

The robot calculates the following master mixes and buffers batch sizes from 8 to 96 samples per run supplied by the operator, aliquoting each into a shared master-mix MIDI plate before the run proper begins:

Master mix / buffer	Key components	Used in
Dephosphorylation Mix	Fast AP (thermosensitive alkaline phosphatase); optional USER enzyme added manually by the operator	Process 1
Adapter/Splint Mix	Biotinylated splint oligonucleotide annealed to the first (partial) adapter strand	Process 2
Ligation Mix I	T4 DNA Ligase HC; 50% PEG-8000	Process 2
Washed C1 Beads	Pre-washed Dynabeads MyOne C1 streptavidin beads	Process 3
Fill-in / Klenow Mix	Klenow Fragment (exo-); 10x Klenow reaction buffer; 2% Tween-20; water	Process 4
Ligation Mix II	T4 DNA Ligase; 10x T4 DNA ligase buffer; 50% PEG-4000	Process 5
Wash buffers	0.1x BWT (bead wash/Tris-Tween), SDS wash buffer, Stringency wash buffer	Processes 3–5
EBT (elution buffer)	Tris-Tween elution/dilution buffer	Process 6

5. Workflow

The method is organized into a preparation step (Process 0) followed by six sequential library-construction processes (Processes 1–6), each announced on-screen and, optionally, by e-mail notification.

Process 0 – Master-mix preparation

Before samples reach each reaction step, the robot aliquots every master mix (dephosphorylation, adapter/splint, Ligation Mix I and II, and fill-in/Klenow mixes) from stock tubes into the master-mix MIDI plate. Where a component must be added by hand (for example, USER enzyme for damage-repair runs), the robot calculates and displays the exact volume required per well and pauses for the operator to pipette it in and confirm before the run continues.

Process 1 – Uracil removal, dephosphorylation, and heat denaturation

Dephosphorylation Mix is dispensed onto the sample plate, which is then transported to the on-deck thermal cycler and incubated at 37 °C for 10 minutes, followed by heat denaturation at 95 °C for 2 minutes. The plate is then moved to a 4 °C cooled position to keep the now single-stranded template stable ahead of ligation.

Process 2 – Splinted ligation of the first adapter

The biotinylated adapter/splint duplex and Ligation Mix I (T4 DNA Ligase HC with PEG-8000) are added to each sample. The reaction is incubated at 37 °C for 1 hour, then heat-treated at 95 °C for 2 minutes to inactivate the ligase and release excess splint oligonucleotide, before the plate is returned to 4 °C.

Process 3 – Immobilization of ligation products on beads

Pre-washed MyOne C1 streptavidin beads are added to each reaction and incubated at room temperature for 30 minutes, with the plate mixed by pipetting every 5 minutes so the biotinylated, ligated molecules bind the beads. The plate then moves to the magnetic separation station (about 10 minutes to collect the beads), the supernatant is removed, and the beads are washed in turn with 0.1x BWT, SDS, and Stringency buffers to remove unligated adapter, protein, and non-specifically bound material.

Process 4 – Second-strand synthesis

With the library molecules still bead-bound, Klenow fill-in mix is added and the plate is incubated on the thermal cycler to synthesize the complementary strand from the 3' end created during ligation. Beads are then returned to the magnetic station, the supernatant is removed, and the beads are washed again (0.1x BWT with SDS, then stringency buffer) before the second ligation.

Process 5 – Ligation of the second adapter

Ligation Mix II (T4 DNA Ligase with PEG-4000) is added directly to the bead-bound, now double-stranded molecules and incubated at room temperature for 1 hour, with mixing by pipetting every 5 minutes, to blunt-end ligate the second adapter onto the newly synthesized strand. The beads are washed once more (0.1x BWT/SDS, then stringency buffer) to remove excess adapter and enzyme.

Process 6 – Elution and 50x dilution

The bead-bound, double-adapter library is heat-eluted on the thermal cycler at 95 °C for 2 minutes — denaturing the streptavidin–biotin interaction and releasing the library into solution — then placed on the magnetic rack to pellet the beads. The eluted supernatant is transferred to a fresh plate, and EBT elution buffer is added to give a 50x dilution, producing a working-concentration library plate.

6. Operator interaction points

At the start of a run, the operator chooses the batch size (8–96 samples), selects which of Processes 1–6 to execute or to treat as mandatory stop points, and can override default incubation temperatures and timers from the values built into the method. Deck-loading dialogs then show, with reference images, which labware (sample plate, MIDI reagent plates, temporary plates, tip racks, troughs, LoBind plates) belongs at each deck position.

During the run, the method pauses for manual steps such as adding USER enzyme to the dephosphorylation mix, and prompts the operator to refill tip racks at set points (tips are otherwise replenished automatically via the Cytomat stacker). If the method is aborted, active devices are shut down and the operator is shown a checklist: empty the tip waste, empty the liquid waste container,

and remove and store reagents and plates. Optional e-mail notifications can be sent at the start and end of each process for remote run monitoring.

7. Output

The method produces a 50x-diluted, plate-based ssDNA library with adapters on both ends but without sample-specific indices. The output plate is ready for indexing PCR and standard library quality control (quantification and fragment-size analysis) prior to pooling and sequencing.

8. Notes

This overview was reconstructed from the Hamilton Venus method files supplied for “MPI_ssDNA_LibraryPreparation” (script, subroutine, step, and compiled method files) and is intended as a workflow-level reference. Exact reagent volumes, deck positions, and labware part numbers are defined in the method files themselves and in the associated lab SOP; these should be consulted directly before running samples or training new users on the platform.